

MNYMO Estimathon 2024

MINNESOTA YOUTH MATH OUTREACH

August 31, 2024

Problems for the estimathon. As a reminder, the score for your team is

$$\left(10 + \sum_{\text{good intervals}} \left\lfloor \frac{\max}{\min} \right\rfloor\right) \cdot 2^{13-(\# \text{ of good intervals})}.$$

All answers are positive reals.

Problem 1. Estimate the population density of penguins in Antarctica, in penguins per square mile.

Problem 2. Estimate the geometric mean of Spotify streams Taylor Swift has on her most popular album and least popular album as of August 21, 2024.

Problem 3. Estimate the sum of the elevation of the 30 mountains in the United States with the greatest topographic elevation, in feet.

Problem 4. Estimate the number of times the phrase “slim shady” occurs in the lyrics of “The Real Slim Shady” by Eminem.

Problem 5. Estimate the area of the quadrilateral not intersecting itself formed by connecting the current home stadiums of the last 4 NFL championship-winning teams, in square miles (disregarding repeat championships).

Problem 6. Estimate the exterior surface area of the Eiffel Tower, in square inches.

Problem 7. Estimate the number of prime numbers less than 1,000,000 (1 million).

Problem 8. Estimate the length of the longest border any country shares with China, in miles.

Problem 9. Estimate the number of silver medals the United States has won in both summer and winter Olympic events since 2000.

Problem 10. Estimate the number of nickels minted in 2023.

Problem 11. Estimate the number of graphing calculators Texas Instruments has sold since 1990.

Problem 12. Estimate the number of views of Cocomelon’s most viewed video on YouTube as of August 21, 2024.

Problem 13. Estimate the value of the quotient when the mean of the answers to all previous problems is divided by the median.